



Monitor & Control (M&C) Interface Control Document (ICD) for the GSFC DTN Bundle Protocol Node (BPNode)

NASA GSFC DTN Project 450.2-DTN-MCICD

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Preface

This document is under configuration management of the Goddard Space Flight Center (GSFC) DTN Project Configuration Control Board (CCB). A complete revision or Document Change Notice (DCN) will be issued to update the document for any future approved changes.

The GSFC DTN Project assumes responsibility for this document and updates it as required to meet its needs. Reviews of this document are performed at least annually, and updates to this document are made when appropriate.

Changes to this document require prior approval of the Change Authority listed on the signature page. Proposed changes shall be submitted to the DTN Configuration Management Engineer (CME) along with supporting material justifying the proposed change. Questions or comments concerning this document should be addressed to: GSFC DTN Systems Engineer or Product Development Lead (PDL).

Change History Log

Revision	Effective Date	Description of Changes (Reference the CCR & CCB/ERB Approval Date)
-	07/12/2024	Initial Release per ESC-CCR-0644 07/12/2024
1	09/30/2025	Build 7.0
2	03/17/2026	Build 7.0.5 updates

Approvals

The signatories below indicate with their signature their commitment to the implementation of this Plan.

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Chapter 1. Scope

1.1. Introduction

The Interface Control Document (ICD) is a controlled document that defines data interfaces and exchange formats between the Delay/Disruption-Tolerant Networking (DTN) Bundle Protocol (BP) nodes and Monitor and Control (M&C) ground systems. BP is a networking protocol that transmits data between nodes even when connectivity is not continuous. It is designed to provide reliable end-to-end communication where other networking protocols may not be effective.

The current Bundle Protocol Node (BPNode) is a DTN Node application that implements the Bundle Protocol version 7 (BPv7). GSFC developed BPNode for a mixed-architecture deployment environment, which means that the node may run on a variety of operating systems, although it has been only tested on Linux so far.

This document is developed to ensure that each side of the interface is correctly designed and compatible.

1.2. Build 7.0.5 Release Notes

Although all the commands, tables, and telemetry packets in this ICD have hooks in the Build 7.0.5 code, some of them do not have the underlying functionality implemented.

The following directives have not been implemented:

- Startup directives
- Policy directives
- reset-source-counters
- add-mib-array-key
- remove-mib-array-key

The following telemetry packets have not been implemented:

- Per-Source MIB Configuration
- Per-Source MIB Counters

The following tables have not been implemented:

- Compressed Reporting
- MIB Source Configuration
- Source Authorization Policy
- Custody Authorization Policy
- Custodian Authorization Policy
- Report-To-EID Authorization Policy

- Source Latency Policy
- Storage

For telemetry packets or tables where only certain fields have not been implemented, these fields will be indicated with *italics*.

1.3. Software Context

The BPNode software runs on Linux and real-time operating systems in a context where BP clients (BPApps from RFC-9171) communicate directly with BPNode. BPNode expects a Publish/Subscribe pattern where a BP app publishes Application Data Units (ADUs) and BPNode subscribes to them. Likewise, a BP app subscribes for expected ADUs and BPNode publishes them. The rest of the diagram below refers to:

1. Core Flight System (cFS) applications, such as
 - a. Telemetry Output (TO)
 - b. Scheduler (SCH)
 - c. Command Ingest (CI)
2. Operating System Abstraction Layer (OSAL)
3. Convergence Layer Adapters (CLAs)
4. Mission Operations Center (MOC), bottom center, which represents the flight controllers and ground systems that communicate with the spacecraft through the M&C interface.

Figure 1. BPNode Context Diagram

1.4. Configuration Management

Interface configuration management is documented in the Configuration Management Plan (450.2-DTN-CMP) baselined in Technical Data Management System (TDMS). DTN CM includes software naming conventions and Discrepancy Reporting management.

1.5. Requirements Verification

Requirements are stored in the GSFC DOORS database and formally exported to TDMS after review and approval by a Configuration Control Board (CCB). The Requirements Verification Traceability Matrix (RVTM) provides verification traceability to testing.

DOORS synchronizes requirements to MagicDraw, which produces system and component diagrams as well as views of requirements that are saved in Teamwork Cloud. DOORS maps requirements to software modules, components, software versions, and source code files.

1.6. Interface Characteristics and Functions

DTN Nodes provide Management Information Base (MIB) objects to collect information required by

Network Management. Mission M&C, DTN Network Management (DTNNM), and MOC use the same MIB-defined objects, which include directives, configuration containers, policy information, events, and telemetry. MIB object transport is mission-defined and has significant mission security, authority, and operational constraints.

DTN nodes support both policy and configuration management interfaces:

- Policy is the management interface that defines user access to network services and resources. It is negotiated between users and service providers to ensure adherence to user Service Level Agreements (SLA). DTNNM applies Node Policy based on protocol parameters and distributes it to node operators. Policy can change per contact schedule. The Internet Engineering Task Force (IETF) and Consultative Committee for Space Data Systems (CCSDS) develop policy distribution format and transport.
- Configuration is a management interface to configure node services and resources. It is more static but can change due to off-nominal operations, fault handling, or policy modifications.

DTNNM ensures that configurations are consistent across the network to support the user SLAs. MIB object scope can address sets of related bundles, all bundles, and node operations. Local and remote MIB object transports are necessary for robust operations and fault handling. All MIB objects have associated Level 6 requirements.

1.7. Space Packets

Once the BPNode starts, the M&C connects to it and establishes distinct connections for the exchange of control directives (command messages) and status messages (telemetry). Both types of messages are standard CCSDS Space Packets as defined in CCSDS 133.0.B-2. The User Data Field contains either the control directive sent by the M&C or the Status Data provided by the BPNode, differentiated by the Packet Type indicator, which is set to 0 for telemetry packets and 1 for telecommands.

Bit 4 of the Packet Primary Header contains the Secondary Header Flag, which indicates the presence or absence of the Packet Secondary Header within this Space Packet. It is set to '0' if a secondary header is absent and '1' if it is present. The latter is the case for both types of DTN packets:

- In a command message, the secondary header contains the Function Code (it is the first data point in the User Data Field) and the command checksum value. It also contains parameters associated with that function, encoded in accordance with their associated data type.
- In a telemetry message, it contains the time stamp.

1.8. DTN-Specific Field Types

Most of the mnemonic field types are standard data types (uint, string, etc), but the following data types will also be referenced in this ICD and defined here:

Table 1. Endpoint ID (EID)

Subfield	Description	Type
Scheme	Defines how to parse scheme-specific part (SSP), is either DTN (1) or IPN (2)	uint64 enum
IpnSspFormat	If the Scheme is IPN, defines whether it is in the 2 or 3 digit format	uint64 enum
Allocator	Unique identifier for controlling organization	uint64
Node	Unique identifier for system that implements DTN communications protocol service	uint64
Service	Unique identifier for DTN communication protocol service	uint64

Table 2. Endpoint ID (EID) Pattern

Subfield	Description	Type
Scheme	Defines how to parse scheme-specific part (SSP), is either DTN (1) or IPN (2)	uint64 enum
IpnSspFormat	If the Scheme is IPN, defines whether it is in the 2 or 3 digit format	uint64 enum
MaxAllocator	Unique identifier for controlling organization, maximum inclusive value	uint64
MinAllocator	Unique identifier for controlling organization, minimum inclusive value	uint64
MaxNode	Unique identifier for system that implements DTN communications protocol service, maximum inclusive value	uint64
MinNode	Unique identifier for system that implements DTN communications protocol service, minimum inclusive value	uint64
MaxService	Unique identifier for DTN communication protocol service, maximum inclusive value	uint64
MinService	Unique identifier for DTN communication protocol service, minimum inclusive value	uint64

Chapter 2. Documentation

2.1. Reference Documents

The following documents serve as reference material for this document.

Table 3. Reference Documents

Document No.	Document Title
RFC-9171	Bundle Protocol Specification (Version 7)
CCSDS 734.20-O-1	CCSDS Bundle Protocol Specification
CCSDS 133.0.B-2	Space Packet Protocol
450.2-DTN-SRD-L6	DTN Level 6 Systems Requirements
CCSDS 734.6-O-1	CCSDS Custody Transfer and Compressed Bundle Status Reporting (Draft Orange Book)

Chapter 3. Directives

DTN subscribes to cFS directive-type CCSDS command messages from the cFS Software Bus (SB). DTN directives are listed below. The Space Packet secondary header for commands specifies the Function Code, which identifies the action being requested of the Node. It is the first data point in the User Data Field, which also contains parameters associated with that function, encoded in accordance with their associated data type.

Table 4. Directives Packet Secondary Header

Name	Description	Type	Start Byte	Start Bit	Length Bits
Reserved	Reserved	uint	6	0	1
FunctionCode	Unique function code for each directive	uint	6	1	7
Checksum	Command checksum value	uint	7	0	8



Directives followed by an asterisk are deferred to a future build (disclaimer: future build items may be subject to change). The subsequent sections discuss each directive in greater detail.

Directive parameters can be formatted as a directive message or table. In general, the issuance of a **valid** control directive increments the **valid** command counter, the issuance of an *invalid* control directive increments the *invalid* command error counter. A directive may be invalid because of the wrong Function Code, length, format, or parameters. The subsequent sections discuss each directive in greater detail.

3.1. Startup Directives

Startup directives add and start all applications and control storage and metadata. None of the startup directives have parameters.

3.1.1. Add All Applications

Name

add-all-applications

Function Code

1

Table

- Channel Configuration Table
- ADU Proxy Table

Parameter(s)

none

Description

Registers with the node all client applications in the channel configuration table.

3.1.2. Start All Applications**Name**

start-all-applications

Function Code

2

Table

- Channel Configuration Table
- ADU Proxy Table

Parameter(s)

none

Description

Starts all applications configured in the channel configuration table. Begins accepting and delivering ADUs for all configured applications.

3.1.3. Verify Bundle Storage***Name**

verify-bundle-storage

Function Code

3

Table

no

Parameter(s)

none

Description

Verifies headers of bundles in persistent storage upon a cold restart.

3.1.4. Initialize Bundle Storage***Name**

initialize-bundle-storage

Function Code

4

Table

no

Parameter(s)

none

Description

Executes on node startup for a cold start and deletes all bundles in storage for a factory reset.

3.1.5. Verify Bundle Metadata***Name**

verify-bundle-metadata

Function Code

5

Table

no

Parameter(s)

none

Description

Checks whether the bundle metadata reflects the actual bundle storage.

3.1.6. Rebuild Bundle Metadata***Name**

rebuild-bundle-metadata

Function Code

6

Table

no

Parameter(s)

none

Description

Rebuilds bundle storage metadata from stored bundles, including their indices (source Endpoint ID (EID), destination EID, next action time).

3.1.7. Clear Volatile***Name**

clear-volatile

Function Code

7

Table

no

Parameter(s)

none

Description

Executes on warm restart and clears volatile data, including bundle metadata and bundles that are not in persistent storage.

3.1.8. Reload Saved Data*

Name

reload-saved-data

Function Code

8

Table

No

Parameter(s)

None

Description

Reload saved node configuration, bundle metadata from persistent storage, and saved MIB counters.

3.2. Counter Directives

These directives instruct the node to reset various counters.

3.2.1. Reset All Counters

Name

reset-all-counters

Function Code

9

Table

No

Parameter(s)

None

Description

Sets all resettable MIB counters to zero.

3.2.2. Reset Counter

Name

reset-counter

Function Code

10

Table

no

Parameter(s)

- uint16 *mibArrayIndex*: source MIB counter array index corresponding to counter (if the counter to reset is a node MIB counter, should be set to the maximum number of source MIB sets allowed, 10 by default)
- uint16 *spare*: spare bytes
- uint32 enum *counter*: Counter to reset. See the Field IDs of the Node and Source MIB Counters Packets for the enumeration values.

Description

Sets to zero the MIB counter specified by the parameter.

3.2.3. Reset Source Counters*

Name

reset-source-counters

Function Code

11

Table

no

Parameter(s)

- uint16 *mibArrayIndex*: source MIB counter array index
- uint16 *spare*: spare bytes

Description

Sets to zero all resettable MIB counters associated with a source EID pattern identified by the parameter.

3.2.4. Reset Bundle Counters

Name

reset-bundle-counters

Function Code

12

Table

no

Parameter(s)

None

Description

Sets all bundle-related counters to zero.

3.2.5. Reset Error Counters

Name

reset-error-counters

Function Code

13

Table

no

Parameter(s)

- uint16 *mibArrayIndex*: source MIB counter array index
- uint16 *spare*: spare bytes

Description

Sets all error counters to zero.

3.3. Application Directives

Application directives add, start, stop, and remove specific applications, as well as set the channel's registration state.

3.3.1. Add Application

Name

add-application

Function Code

14

Table

- Channel Configuration Table
- ADU Proxy Table

Parameter(s)

uint32 *ChanId*: The index in the Channel Configuration Table with this application's configurations, also functions as a unique identifier for this application.

Description

This directive adds a new application by:

- Setting the channel configuration based on client application configuration
- Establishing mapping between client application connection and channel
- Opening an ADU channel

3.3.2. Remove Application

Name

remove-application

Function Code

15

Table

no

Parameter(s)

uint32 *ChanId*: The index in the Channel Configuration Table with this application's configurations, also functions as a unique identifier for this application.

Description

Terminates the connection, closes the ADU channel, and flushes the egress queue for the application specified by the parameter.

3.3.3. Set Registration State

Name

set-registration-state

Function Code

16

Table

- Channel Configuration Table
- ADU Proxy Table

Parameter(s)

- uint32 *ChanId*: The index in the Channel Configuration Table with this application's configurations, also functions as a unique identifier for this application
- uint8 enum *registrationState*: active (0), passive-defer (1), or passive-abandon (2)

Description

Sets given application's registration state to specified state. If a channel is set to active, ADUs will be delivered automatically when the channel is started. If a channel is set to passive-defer, ADUs will be stored instead of delivered, even if the channel is started. If a channel is set to passive-abandon, ADUs will be deleted and discarded instead of delivered.

3.3.4. Start Application

Name

start-application

Function Code

17

Table

no

Parameter(s)

uint32 *ChanId*: The index in the Channel Configuration Table with this application's configurations, also functions as a unique identifier for this application.

Description

Verifies the channel configuration of the application indicated by the parameter and begins moving payloads bidirectionally between the node and the given application.

3.3.5. Stop Application

Name

stop-application

Function Code

18

Table

no

Parameter(s)

uint32 *ChanId*: The index in the Channel Configuration Table with this application's configurations, also functions as a unique identifier for this application.

Description

Stops moving bundles bidirectionally between the node and the given application.

3.4. Policy Directives

Policy Directives add and remove authorized EIDs.



None of the functionality of the policy directives has been implemented as of build 7.0.5

3.4.1. Add Authorized Sources*

Name

add-authorized-sources

Function Code

19

Table

Yes

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Adds the EID pattern to a set of authorized source EIDs if the pattern being added does not exceed the maximum size of authorized sources.

3.4.2. Remove Authorized Sources*

Name

remove-authorized-sources

Function Code

20

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Removes the EID pattern from a set of authorized source EIDs.

3.4.3. Add Authorized Custody Sources*

Name

add-authorized-custody-sources

Function Code

21

Table

Yes

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Adds the EID pattern to a set of authorized custody source EIDs.

3.4.4. Remove Authorized Custody Sources*

Name

remove-authorized-custody-sources

Function Code

22

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Removes the EID pattern from a set of authorized custody source EIDs.

3.4.5. Add Authorized Custodians*

Name

add-authorized-custodians

Function Code

23

Table

Yes

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Adds the EID pattern to a set of authorized custodian EIDs.

3.4.6. Remove Authorized Custodians*

Name

remove-authorized-custodians

Function Code

24

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Removes the EID pattern from a set of authorized custodian EIDs.

3.4.7. Add Authorized Report-to EID*

Name

add-authorized-report-to-eid

Function Code

25

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Adds the EID pattern to a set of authorized report-to EIDs.

3.4.8. Remove Authorized Report-to EID*

Name

remove-authorized-report-to-eid

Function Code

26

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Removes the EID pattern from a set of authorized report-to EIDs.

3.4.9. Add Latency*

Name

add-latency

Function Code

27

Table

No Parameter(s)

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Adds the source EID pattern and latency information to the set of source latency policies.

3.4.10. Remove Latency*

Name

remove-latency

Function Code

28

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Removes source EID pattern and latency information from the set of source latency priorities.

3.5. Contact Directives

Contact directives set up, start, stop, and teardown CLA contacts. Note that in build 7.0.5, only the UDP CLA and the SB CLA are supported.

3.5.1. Set Up

Name

contact-setup

Function Code

29

Table

Contact Table

Parameter(s)

uint32 *contactId*: unique CLA contact ID, points to row in Contact Table with necessary configurations

Description

Establishes and configures CLA to obtain output queue bundles from Node storage to send to Convergence Layer (CL). It creates the output queue and configures the rate at which CLA sends and receives bundles to and from CL. A sequence ID is assigned for all custodial bundles that come in on this contact. If there are any stored custodial bundles that can be egressed on this contact, their retransmission timers are updated to that of this contact. If CLA type is:

- LTP - the directive also configures the destination LTP engine ID
- TCPCLP - it also establishes a session with peer entity.



This directive is rejected if setting up the contact would exceed the allowed maximum of simultaneous contacts.

3.5.2. Start

Name

contact-start

ID

30

Table

No

Parameter(s)

uint32 *contactId*: unique CLA contact ID

Description

Starts transferring bundles between the underlying network and the node.

3.5.3. Stop

Name

contact-stop

ID

31

Table

No

Parameter(s)

uint32 *contactId*: unique CLA contact ID

Description

This directive performs the following:

- stops transferring bundles to and from CL
- requests the CL to cancel the transfers in progress
- sends any Compressed Custody Signals (CCSs) currently under construction

If CL is LTP, the directive requests LTP to cancel active session(s) and notify the node of complete and incomplete bundle transmissions.

3.5.4. Tear Down

Name

contact-teardown

ID

32

Table

No

Parameter(s)

uint32 *contactId*: unique CLA contact ID

Description

Disestablishes CLA, frees all CLA resources, stores any bundles remaining in the egress queue, and deletes custody timers.

3.6. MIB Directives

MIB directives modify the MIB configurations.

3.6.1. Add MIB Array Key*

Name

add-MIB-array-key

Function Code

33

Table

Yes

Parameter(s)

EID Pattern[4] *eidPatterns*: EID patterns to add

Description

Adds the given EID patterns as key to the map of MIB configuration elements and counters accessed by the source EID.

3.6.2. Remove MIB Array Key*

Name

remove-MIB-array-key

Function Code

34

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Removes the elements indexed by given EID pattern from the map of MIB configuration elements and counters accessed by source EID.

3.6.3. Set MIB Item

Name

set-MIB-item

Function Code

35

Table

Yes

Parameter(s)

- EID Pattern *eid*: EID(s) for which to set the MIB item. Only the node MIB is supported for build 7.0.5.
- uint32 enum *mibId*: MIB item ID. See the Field IDs of the Node and Source MIB Configuration Packets/Tables for the enumeration values.
- uint32 *value*: value to which the MIB item will be set

Description

Sets the value of a MIB configuration item specified by the parameter.

3.7. Storage Directives

Storage directives add and remove storage allocation.

3.7.1. Add Storage Allocation*

Name

add-storage-allocation

Function Code

36

Table

Yes Parameter(s)

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Adds the storage partition of a specified size for storing bundles whose source EID matches the given pattern.

3.7.2. Remove Storage Allocation*

Name

remove-storage-allocation

Function Code

37

Table

No

Parameter(s)

uint32 *Placeholder*: Placeholder parameter

Description

Removes the storage partition corresponding to a given EID pattern.

3.7.3. Perform Self-Test*

Name

perform-self-test

Function Code

38

Table

No

Parameter(s)

None

Description

Perform TBD tests, returning pass/fail.

3.7.4. Clean Storage

Name

clean-storage

Function Code

39

Table

No

Parameter(s)

None

Description

Perform any needed storage cleanup, including database defragmentation. This operation is run automatically anytime the node detects that the stored bundle count has dropped to 0.

3.8. Routine Directives

Routine directives test aliveness, trigger the node to wake up, and send telemetry packets.

3.8.1. Wakeup

Name

Wakeup

Function Code

39

Table

No

Parameter(s)

None

Description

Triggers BPNode tasks to wake up and start processing directives and bundles, as well as

performing maintenance activities, such as time calculation, framework management, and garbage collection.

3.8.2. Send Node MIB Configuration HK

Name

send-node-mib-config-hk

Function Code

40

Table

No

Parameter(s)

None

Description

Sends the Per Node MIB Configuration telemetry packet.

3.8.3. Send Per Source MIB Configuration HK

Name

send-per-source-mib-config-hk

Function Code

41

Table

No

Parameter(s)

None

Description

Sends the Per Source MIB Configuration telemetry packet.

3.8.4. Send Node MIB Counters HK

Name

send-node-mib-counters-hk

Function Code

42

Table

No

Parameter(s)

None

Description

Sends the Per Node MIB Counters telemetry packet.

3.8.5. Send Per Source MIB Counters HK**Name**

send-per-source-mib-counters-hk

Function Code

43

Table

No

Parameter(s)

None

Description

Sends the Per Source MIB Counters telemetry packet.

3.8.6. Send Storage HK**Name**

send-storage-hk

Function Code

44

Table

No

Parameter(s)

None

Description

Sends the Storage telemetry packet.

3.8.7. Send Channel/Contact Status HK**Name**

send-channel-contact-status-hk

Function Code

45

Table

No

Parameter(s)

None

Description

Sends the Channel/Contact Status telemetry packet.

3.8.8. Send Node MIB Reports HK

Name

send-node-mib-reports-hk

Function Code

46

Table

No

Parameter(s)

None

Description

Sends the Per Node MIB Reports telemetry packet.

3.8.9. Noop

Name

Noop

Function Code

0

Table

No

Parameter(s)

None

Description

Ensures that the connection is still alive. The BPNode version information is sent out in an event message.

Chapter 4. Telemetry

BPNode converts telemetry data received from its modules from internal library format to standard CCSDS Space Packets and publishes them to cFS SB to send them to M&C. The DTN Node generates the following telemetry packets:

- Node MIB Configuration
- Per-Source MIB Configuration
- Node MIB Counters
- Per-Source MIB Counters
- Node MIB Reports
- Storage HK
- Channel and Contact Status HK.

The Space Packet secondary header for telemetry indicates host time (absolute time value in an epoch defined by cFS) in seconds and subseconds.

Table 5. Telemetry Packet Secondary Header

Name	Description	Type	Start Byte	Start Bit	Length Bits
Seconds	Seconds since host epoch	uint	6	0	32
Subseconds	Subseconds (one subsecond = 2^{-16} seconds)	uint	8	0	16

In addition, DTN also applies timestamps within the body of generated telemetry packets so that operators can determine the DTN time:

Table 6. DTN Time Terminology

Term	Data Type	Units	Description
currentMonotonicTime	uint64	msec	The current time indicated by the monotonic clock, usually the time elapsed since the clock was last powered on
currentCorrelationFactor or	uint64	msec	The offset needed to correlate the current monotonic time to the DTN epoch (DTN Time = currentMonotonicTime + currentCorrelationFactor)

Upon receiving one of the send HK directives, the node generates a CCSDS space packet, adds the relevant telemetry fields, acquires time data, and sends the packet to SB.

The telemetry packets generated by the node are listed in the following sections.



Although the per-source packets are generated by BPNode, in build 7.0.5 none of the fields are implemented, so they will always be set to 0. For the remaining packets, any fields in *italics* are similarly not implemented and will always be set to 0. (Disclaimer: future build items may be subject to change.)

4.1. Node MIB Configuration

This telemetry packet contains all node MIB configurations variables.

Table 7. MIB Node Configuration

Field ID	Field Name	Field Type	Description
N/A	instanceEID	EID	Endpoint ID of this node
0	<i>paramBundleSizeNoFragment</i>	uint32	Maximum size of bundles that can traverse DTN without additional bundle layer fragmentation (provided by DTNNM).
1	paramSetMaxSequenceNum	uint32	Maximum bundle sequence number.
2	paramSetMaxPayloadLength	uint32	Maximum payload length for fragmentation.
3	paramSetMaxBundleLength	uint32	Maximum bundle length.
4	paramSupportCustody	uint32	Whether or not to support custody transfer operations on this node (0=false, 1=true)
5	paramSetMaxLifetime	uint32	Maximum bundle lifetime allowed on node

4.2. Per-Source MIB Configuration*

This telemetry packet contains the MIB configuration variables for each set of sources.



None of the fields in this packet are implemented.

Field Name	Data Type	# Entries	Description
sourceConfigs	Source Configs Array	10 by default	Configurations for each source (see table below)

The Source Configurations Array contains the following fields for each entry:

Table 8. Source Configuration Array

Field ID	Field Name	Data Type	# Entries	Description
N/A	SrcEIDs	EID Pattern	Max MIB Per Source EID Patterns, 4 by default	Source EID Patterns
5	ParamSetMaxLifetime	uint32	1	Maximum bundle lifetime (in seconds) that ensures that bundle retention until its expiration time will not degrade operation of the receiving node. Used by node to determine when the bundle must be deleted to prevent network performance degradation.
6	ParamSetMaxBSRGenerationRate	uint32	1	Maximum number of BSRs per minute that can be generated for each source
7	ParamSetMaxCBRGenerationRate	uint32	1	Maximum number of CBRs per minute that can be generated for each source
8	BundleSetBehaviorReceivedBSRGenerate	uint32	1	Flag indicating bundle reception status reports should be generated
9	BundleSetBehaviorAcceptedBSRGenerate	uint32	1	Flag indicating bundle custody accepted status reports should be generated
10	BundleSetBehaviorForwardedBSRGenerate	uint32	1	Flag indicating bundle forwarded status reports should be generated
11	BundleSetBehaviorDeliveredBSRGenerate	uint32	1	Flag indicating bundle delivered status reports should be generated
12	BundleSetBehaviorDeletedBSRGenerate	uint32	1	Flag indicating bundle deleted status reports should be generated

Field ID	Field Name	Data Type	# Entries	Description
13	BundleSetBehaviorReceivedCBRGenerate	uint32	1	Flag indicating bundle reception status reports should be generated
14	BundleSetBehaviorAcceptedCBRGenerate	uint32	1	Flag indicating bundle custody accepted status reports should be generated
15	BundleSetBehaviorForwardedCBRGenerate	uint32	1	Flag indicating bundle forwarded status reports should be generated
16	BundleSetBehaviorDeliveredCBRGenerate	uint32	1	Flag indicating bundle delivered status reports should be generated
17	BundleSetBehaviorDeletedCBRGenerate	uint32	1	Flag indicating bundle deleted status reports should be generated
N/A	Spare	uint32	1	Spare bytes for padding

4.3. Node MIB Counters

This telemetry packet includes all node MIB counter variables.

Table 9. Node MIB Counters

Field ID	Field Name	Data Type	Description
0	aduCountDelivered	uint32	Number of ADUs delivered to an application.
1	aduCountReceived	uint32	Number of ADUs received
2	bundleCountAbandoned	uint32	Number of abandoned bundle payloads
3	bundleCountCustodyRejected	uint32	Number of unsuccessful custody transfers from the local node to the neighboring node
4	bundleCountCustodyRequest	uint32	Number of bundles requesting custody transfer

Field ID	Field Name	Data Type	Description
5	bundleCountCustodyReforwarded	uint32	Number of bundles reforwarded due to custody timeout
6	bundleCountCustodyTransferred	uint32	Number of successful custody from the local node to the neighboring node
7	bundleCountDeleted	uint32	Total number of bundle deletions
8	bundleCountDeletedExpired	uint32	Number of bundles deleted due to Lifetime Expired condition
9	bundleCountDeletedHopExceeded	uint32	Number of bundles deleted due to Hop Limit Exceeded condition
10	bundleCountDeletedNoStorage	uint32	Number of bundles deleted due to insufficient storage
11	bundleCountDeletedTooLong	uint32	Number of bundles deleted due to being longer than the maximum bundle length
12	<i>bundleCountDeletedTrafficPared</i>	uint32	Number of bundles deleted due to Traffic Pared condition
13	<i>bundleCountDeletedUnauthenticated</i>	uint32	Number of bundles deleted due to unrecognized source EID
14	bundleCountDeletedUnintelligible	uint32	Number of bundles deleted due to Block Unintelligible condition
15	bundleCountDeletedUnsupportedBlock	uint32	Number of bundles deleted due to Unsupported Block condition
16	bundleCountDelivered	uint32	Number of bundles delivered
17	bundleCountDepleted	uint32	Number of bundles whose custody transfer was rejected due to lack of storage
18	bundleCountDiscarded	uint32	Number of bundles discarded
19	bundleCountForwarded	uint32	Number of bundles forwarded

Field ID	Field Name	Data Type	Description
20	<i>bundleCountForwardedFailed</i>	uint32	Number of bundles where forwarding failed
21	<i>bundleCountFragmented</i>	uint32	Number of bundles that needed fragmentation
22	<i>bundleCountFragmentError</i>	uint32	Number of fragments discarded due to error
23	bundleCountGeneratedAccepted	uint32	Number of accepted bundle transmission requests
24	bundleCountGeneratedCustodySignal	uint32	Number of bundles for which custody signals were generated
25	<i>bundleCountGeneratedFragment</i>	uint32	Number of generated bundle fragments
26	bundleCountGeneratedRejected	uint32	Number of rejected bundle transmission requests
27	<i>bundleCountMaxBsrRateExceeded</i>	uint32	Number of BSR bundles not sent due to rate limit
28	<i>bundleCountNoContact</i>	uint32	Number of bundles whose rejected Custody Signals indicated destination is not reachable before expiration
29	bundleCountAcceptedCustody	uint32	Number of bundles whose custody was accepted by the local node
30	<i>bundleCountNoRoute</i>	uint32	Number of bundles whose rejected Custody Signals indicated the destination is not reachable
31	<i>bundleCountReassembled</i>	uint32	Number of bundles delivered that were reassembled fragments
32	bundleCountReceived	uint32	Number of bundles received
33	bundleCountReceivedAdminRecord	uint32	Number of administrative records received

Field ID	Field Name	Data Type	Description
34	<i>bundleCountReceivedBsrAccepted</i>	uint32	Number of "Bundle Accepted" BSRs received
35	<i>bundleCountReceivedBsrDeleted</i>	uint32	Number of "Bundle Deleted" BSRs received
36	<i>bundleCountReceivedBsrDelivered</i>	uint32	Number of "Bundle Delivered" BSRs received
37	<i>bundleCountReceivedBsrForwarded</i>	uint32	Number of "Bundle Forwarded" BSRs received
38	<i>bundleCountReceivedBsrReceived</i>	uint32	Number of "Bundle Received" BSRs received
39	<i>bundleCountReceivedCrsAccepted</i>	uint32	Number of "Bundle Accepted" CRSs received
40	<i>bundleCountReceivedCrsDeleted</i>	uint32	Number of "Bundle Deleted" CRSs received
41	<i>bundleCountReceivedCrsDelivered</i>	uint32	Number of "Bundle Delivered" CRSs received
42	<i>bundleCountReceivedCrsForwarded</i>	uint32	Number of "Bundle Forwarded" CRSs received
43	<i>bundleCountReceivedCrsReceived</i>	uint32	Number of "Bundle Received" CRSs received
44	bundleCountReceivedCustodySignal	uint32	Number of bundles for which custody signals were received
45	<i>bundleCountReceivedFragment</i>	uint32	Number of fragment bundles received
46	bundleCountRedundant	uint32	Number of bundles where successful custody signals were generated for duplicate bundles
47	bundleCountRejectedCustody	uint32	Number of bundles whose custody the local node rejected

Field ID	Field Name	Data Type	Description
48	<i>bundleCountUnintelligibleBlock</i>	uint32	Number of bundles for which Custody Signals indicated the bundle contained an unknown block type
49	<i>bundleCountUnintelligibleEid</i>	uint32	Number of bundles rejected for unknown EIDs
50	bundleCountUnprocessedBlocks	uint32	Number of unprocessed blocks removed
51	bundleAgentAcceptedDirectiveCount	uint32	Number of accepted control directives received from the M&C interface.
52	bundleAgentRejectedDirectiveCount	uint32	Number of rejected invalid control directives received from the M&C interface.
53	<i>bundleCountCcsReceived</i>	uint32	Number of Compressed Custody Signal bundles received.
54	<i>bundleCountGeneratedBsrAccepted</i>	uint32	Number of BSRs of bundle accepted for custody.
55	<i>bundleCountGeneratedBsrDeleted</i>	uint32	Number of BSRs of bundles deleted.
56	<i>bundleCountGeneratedBsrDelivered</i>	uint32	Number of BSRs of bundles delivered.
57	<i>bundleCountGeneratedBsrForwarded</i>	uint32	Number of BSRs of bundles forwarded.
58	<i>bundleCountGeneratedBsrReceived</i>	uint32	Number of BSRs of bundles received generated.
59	<i>bundleCountGeneratedCrs</i>	uint32	Number of CRSs generated since last counter reset.
60	<i>bundleCountGeneratedCrsAccepted</i>	uint32	Number of accepted bundle reports in each CRS.
61	<i>bundleCountGeneratedCrsDeleted</i>	uint32	Number of deleted bundle reports in each CRS.

Field ID	Field Name	Data Type	Description
62	<i>bundleCountGeneratedCrsDelivered</i>	uint32	Number of delivered bundle reports in each CRS.
63	<i>bundleCountGeneratedCrsForwarded</i>	uint32	Number of forwarded bundle reports in each CRS.
64	<i>bundleCountGeneratedCrsReceived</i>	uint32	Number of received bundle reports in each CRS.
65	<i>bundleCountGeneratedCcs</i>	uint32	Number of Compressed Custody Signal bundles generated.
66	<i>bundleCountMaxCrsRateExceeded</i>	uint32	Number of CRS bundles not sent to avoid exceeding maximum rate.
67	<i>bundleCountReceivedCrs</i>	uint32	Number of CRSs received since last counter reset.

4.4. Per-Source MIB Counters

This telemetry packet includes all MIB counter variables for each set of sources.



None of the fields in this packet are implemented.

Table 10. Source Counters*

Field Name	Data Type	# Entries	Description
sourceCounters	Source Counters Array	10 by default	Counters for each source (see table below)

The Source Counters Array contains the following fields for each entry:

Table 11. Source MIB Counters

Field ID	Field Name	Data Type	Description
N/A	sourceEIDs	Array of 4 EID patterns	Source EID patterns for this entry
N/A	activeKeys	uint8	Number of keys are active in the EID patterns
N/A	spare	uint8[3]	Spare for alignment
0	aduCountDelivered	uint32	Number of ADUs delivered to an application.

Field ID	Field Name	Data Type	Description
1	aduCountReceived	uint32	Number of ADUs received
2	bundleCountAbandoned	uint32	Number of abandoned bundle payloads
3	bundleCountCustodyRejected	uint32	Number of unsuccessful custody transfers from the local node to the neighboring node
4	bundleCountCustodyRequest	uint32	Number of bundles requesting custody transfer
5	bundleCountCustodyReforwarded	uint32	Number of bundles reforwarded due to custody timeout
6	bundleCountCustodyTransferred	uint32	Number of successful custody from the local node to the neighboring node
7	bundleCountDeleted	uint32	Total number of bundle deletions
8	bundleCountDeletedExpired	uint32	Number of bundles deleted due to Lifetime Expired condition
9	bundleCountDeletedHopExceeded	uint32	Number of bundles deleted due to Hop Limit Exceeded condition
10	bundleCountDeletedNoStorage	uint32	Number of bundles deleted due to insufficient storage
11	bundleCountDeletedTooLong	uint32	Number of bundles deleted due to being longer than the maximum bundle length
12	<i>bundleCountDeletedTrafficPared</i>	uint32	Number of bundles deleted due to Traffic Pared condition
13	<i>bundleCountDeletedUnauthorized</i>	uint32	Number of bundles deleted due to unrecognized source EID
14	bundleCountDeletedUnintelligible	uint32	Number of bundles deleted due to Block Unintelligible condition

Field ID	Field Name	Data Type	Description
15	bundleCountDeletedUnsupportedBlock	uint32	Number of bundles deleted due to Unsupported Block condition
16	bundleCountDelivered	uint32	Number of bundles delivered
17	bundleCountDepleted	uint32	Number of bundles whose custody transfer was rejected due to lack of storage
18	bundleCountDiscarded	uint32	Number of bundles discarded
19	bundleCountForwarded	uint32	Number of bundles forwarded
20	<i>bundleCountForwardedFailed</i>	uint32	Number of bundles where forwarding failed
21	<i>bundleCountFragmented</i>	uint32	Number of bundles that needed fragmentation
22	<i>bundleCountFragmentError</i>	uint32	Number of fragments discarded due to error
23	bundleCountGeneratedAccepted	uint32	Number of accepted bundle transmission requests
24	bundleCountGeneratedCustodySignal	uint32	Number of bundles for which custody signals were generated
25	<i>bundleCountGeneratedFragment</i>	uint32	Number of generated bundle fragments
26	bundleCountGeneratedRejected	uint32	Number of rejected bundle transmission requests
27	<i>bundleCountMaxBsrRateExceeded</i>	uint32	Number of BSR bundles not sent due to rate limit
28	<i>bundleCountNoContact</i>	uint32	Number of bundles whose rejected Custody Signals indicated destination is not reachable before expiration
29	bundleCountAcceptedCustody	uint32	Number of bundles whose custody was accepted by the local node

Field ID	Field Name	Data Type	Description
30	<i>bundleCountNoRoute</i>	uint32	Number of bundles whose rejected Custody Signals indicated the destination is not reachable
31	<i>bundleCountReassembled</i>	uint32	Number of bundles delivered that were reassembled fragments
32	<i>bundleCountReceived</i>	uint32	Number of bundles received
33	<i>bundleCountReceivedAdminRecord</i>	uint32	Number of administrative records received
34	<i>bundleCountReceivedBsrAccepted</i>	uint32	Number of "Bundle Accepted" BSRs received
35	<i>bundleCountReceivedBsrDeleted</i>	uint32	Number of "Bundle Deleted" BSRs received
36	<i>bundleCountReceivedBsrDelivered</i>	uint32	Number of "Bundle Delivered" BSRs received
37	<i>bundleCountReceivedBsrForwarded</i>	uint32	Number of "Bundle Forwarded" BSRs received
38	<i>bundleCountReceivedBsrReceived</i>	uint32	Number of "Bundle Received" BSRs received
39	<i>bundleCountReceivedCrsAccepted</i>	uint32	Number of "Bundle Accepted" CRSs received
40	<i>bundleCountReceivedCrsDeleted</i>	uint32	Number of "Bundle Deleted" CRSs received
41	<i>bundleCountReceivedCrsDelivered</i>	uint32	Number of "Bundle Delivered" CRSs received
42	<i>bundleCountReceivedCrsForwarded</i>	uint32	Number of "Bundle Forwarded" CRSs received
43	<i>bundleCountReceivedCrsReceived</i>	uint32	Number of "Bundle Received" CRSs received

Field ID	Field Name	Data Type	Description
44	bundleCountReceivedCustodySignal	uint32	Number of bundles for which custody signals were received
45	<i>bundleCountReceivedFragment</i>	uint32	Number of fragment bundles received
46	bundleCountRedundant	uint32	Number of bundles where successful custody signals were generated for duplicate bundles
47	bundleCountRejectedCustody	uint32	Number of bundles whose custody the local node rejected
48	<i>bundleCountUnintelligibleBlock</i>	uint32	Number of bundles for which Custody Signals indicated the bundle contained an unknown block type
49	<i>bundleCountUnintelligibleEid</i>	uint32	Number of bundles rejected for unknown EIDs
50	bundleCountUnprocessedBlocks	uint32	Number of unprocessed blocks removed

4.5. Node MIB Reports

This telemetry packet contains all node MIB reports variables.

Table 12. Node MIB Reports

Field Name	Data Type	Description
systemNodeName	char[32]	Textual name of the entity with DTN functionality. This name is human readable and used to unambiguously identify a node in the network.
systemNodeOwner	char[32]	Textual identifier for the primary manager of the node, who allocates node resources or functions.
systemSoftwareExec	char[32]	Textual identification of the underlying operating system or executive that controls the resources upon which the DTN functionality is running.
systemSoftwareExecVersion	char[32]	Textual representation of the version and patch-level of the software defined via nodeExecutive. Need to know the full OS with version and patch numbers.
bundleAgentSoftwareVersion	char[32]	Version of the BPA

Field Name	Data Type	Description
<i>bundleAgentOperationalState</i>	char[32]	Operational state of the BPA
<i>bundleAgentConfiguration</i>	char[32]	Indication of the BPA configuration
paramSupportedCLAs	char[32]	List of supported CLAs
<i>nodeActiveEndpoints</i>	char[32]	List of active endpoints
BundleIngressRateBitsPerSec	uint32	Rate of bundles received from CLAs in bits per second
BundleIngressRateBundlesPerSec	uint32	Rate of bundles received from CLAs in bundles per second
BundleEgressRateBitsPerSec	uint32	Rate of bundles forwarded to CLAs in bits per second
BundleEgressRateBundlesPerSec	uint32	Rate of bundles forwarded to CLAs in bundles per second
BundleIngestedRateBitsPerSec	uint32	Rate of ADUs received locally in bits per second
BundleIngestedRateBundlesPerSec	uint32	Rate of ADUs received locally in bundles per second
BundleDeliveryRateBitsPerSec	uint32	Rate of bundles delivered locally in bits per second
BundleDeliveryRateBundlesPerSec	uint32	Rate of bundles delivered locally in bundles per second
BundleIngressRejectedRateBitsPerSec	uint32	Rate of bundles received and rejected from CLAs in bits per second
BundleIngressRejectedRateBundlesPerSec	uint32	Rate of bundles received and rejected from CLAs in bundles per second
bundleAgentCtdbSize	uint32	Size in bytes of the Custody Transfer Database
systemNodeUpTime	uint32	Time in seconds since node has been reinitialized
bundleAgentAvailableStorage	uint32	Kilobytes of memory initially allocated for bundle storage.
kbytesCountStorageAvailable	uint32	Kilobytes of storage that is free
bundleCountStored	uint32	Number of bundles currently in storage
bundleCountInCustody	uint32	Number of bundles currently in custody
nodeStartupCounter	uint32	Number of times node has started up

4.6. Storage Telemetry

This packet contains telemetry values related to storage or memory usage.

Table 13. Storage Telemetry

Field Name	Data Type	Description
BytesMemInUse	size_t	Bytes of memory currently in use
BytesMemFree	size_t	Bytes of memory that are free
BytesMemHighWater	size_t	Memory high water mark in bytes
KbStorageInUse	size_t	Kilobytes of storage currently in use
KbBundlesInStor	size_t	Kilobytes of storage currently occupied by bundles

4.7. Channel/Contact Status Telemetry

The telemetry packet contains the following components:

Table 14. Channel/Contact Status Telemetry

Field Name	Data Type	# Entries	Description
channelStatus	Channel Status Array	2 by default	Status for each channel (see table below)
contactStatus	Contact Status Array	1 by default	Status for each contact (see table below)

4.7.1. Channel Status Data

The Channel Status Array contains the following fields for each entry:

Table 15. Channel Status Data

Field Name	Data Type	Description
localServiceNum	uint32	Service number for the application sending/receiving ADUs on this channel
state	uint32 enum	Removed (0), Stopped (1), Added (2), or Started (3)
registrationState	uint32 enum	Active (0), Passive-Defer (1), or Passive-Abandon (2)
spare	uint32	Spare for alignment

4.7.2. Contact Status Data

The Contact Status Array contains the following fields for each entry:

Table 16. Contact Status Data

Field Name	Data Type	Description
state	uint32 enum	Torndown (0), Setup (1), Started (2), or Stopped (3)
spare	uint32	Spare for alignment
EIDs	Array of 3 EID patterns	List of destination EIDs for this contact

Chapter 5. Tables

BPNode contains a built-in table translator and reader of cFS tables. The node receives table parameters through cFE Table Service, parses, translates, and passes them to the relevant internal components, e.g., for node configuration. The tables can be modified by commands to cFE Table Services and/or the specified directive(s). BPNode has the following tables and directives for their implementation:

Table 17. BPNode Tables

Table Name	Directive or Ground Table Load
Compressed Reporting	Ground Table Load
Channel Configuration	Ground Table Load
ADU Proxy Configuration	Ground Table Load
Contact Configuration	Ground Table Load
MIB Configuration per Node	<code>set-MIB-item</code>
MIB Configuration per Source	<code>set-MIB-item</code> , <code>add/remove-MIB-array-key</code>
Storage	<code>add/remove-storage-allocation</code>
Source Authorization Policy	<code>add/remove-authorized-sources</code>
Custody Authorization Policy	<code>add/remove-authorized-custody-sources</code>
Custodian Authorization Policy	<code>add/remove-authorized-custodians</code>
Report-To-EID Authorization Policy	<code>add/remove-authorized-report-to-eid</code>
Source Latency Policy	<code>add/remove-latency</code>



Build 7.0.5 implements only the MIB configuration per node, ADU proxy, channel configuration, and contact configuration tables. Non-implemented fields in those tables are *italicized*. The remaining tables are part of BPNode but not yet used, these are marked with an asterisk. (Disclaimer: future build items may be subject to change.)

5.1. Compressed Reporting*

This table can be updated by a ground table load. A `contact-stop` directive as well as CRS time or size triggers will prompt the processor to send an in-progress CRS. This table's functionality has not been implemented as of build 7.0.5 This table contains the following components:

Table 18. CRS Trigger Data

Field Name	Data Type	# Entries	Description
crsTriggerData	CRS Trigger Array	10 by default	Configurations for each CRS trigger

Table 19. CRS Trigger Data

Field Name	Field Type	Description
destinationEID	EID	Destination EID.
timeTrigger	uint32	A timeout value that triggers sending a CRS after no longer than the specified value.
sizeTrigger	uint32	Maximum CSR size before it is encoded and sent.

5.2. Channel Configuration

This table captures the configurations for flowing bundles to a client application. After loading the table through cFE Table Services, channel configurations in the table can be added or removed from the node with add-application/remove-application directives. The table contains the following components:

Table 20. Channel Configuration

Field Name	Data Type	# Entries	Description
channelConfigs	Channel Configs Array	Maximum number of channels, 2 by default	Configurations for each channel (see table below)

The Channel Configurations Array contains the following fields for each entry:

Table 21. Channel Configuration Array Entry Fields

Field Name	Data Type	Description
AddAutomatically	bool	Load this configuration upon node startup.
AduWrapping	bool	Whether to wrap an ADU in a CCSDS header upon egress.
<i>AduUnwrapping</i>	bool	Whether to unwrap an ADU in a CCSDS header upon ingress.
RegistrationState	uint8 enum	Active (0), Passive-Defer (1), or Passive-Abandon (2).
HopLimit	uint8	Maximum number of forwards/hops.
CrcType	uint8 enum	Primary block CRC type: None (0), CRC-16 (1), or CRC-32 (2).
Spare	uint16	Spare for alignment
IngressBitsPerCycle	size_t	Maximum number of bits to ingress per wakeup cycle. Note: default wakeup rate is 10Hz.
EgressBitsPerCycle	size_t	Maximum number of bits to egress per wakeup cycle. Note: default wakeup rate is 10Hz.
LocalServiceNumber	uint32	Local service number (node number assumed).
BundleProcessingCtrl Flags	uint64	RFC-9171 Bundle Processing Control Flags
Lifetime	uint64	Lifetime for all bundles in milliseconds.

Field Name	Data Type	Description
DestEID	EID	Destination EID
ReportToEID	EID	EID to send status reports to.
PrevNodeBlkConfig	Canonical Block Config	Configurations for the previous node block.
AgeBlkConfig	Canonical Block Config	Configurations for the age block.
HopCountBlkConfig	Canonical Block Config	Configurations for the hop count block.
HopCountBlkConfig	Canonical Block Config	Configurations for the custody transfer block.
PayloadBlkConfig	Canonical Block Config	Configurations for the payload block.

Table 22. Canonical Block Configuration

Field Name	Field Type	Description
includeBlock	bool	Whether to include this block (must be true for the payload block)
CrcType	uint8 enum	CRC type: None (0), CRC-16 (1), or CRC-32 (2).
Spare	uint16	Spare for alignment
BlockNum	uint32	Block number
BlockProcessingCtrlFlags	uint64	Canonical block processing control flags

5.3. ADU Proxy Configuration

This table is the only table not managed by BPLib since it is specific to cFS. It defines the cFS-specific configurations for each ADU Proxy channel. The table contains the following components:

Table 23. ADU Configuration

Field Name	Data Type	# Entries	Description
aduConfigs	ADU Configs Array	Maximum number of channels, 2 by default	Configurations for each channel (see table below)

The ADU Configurations Array contains the following fields for each entry:

Table 24. ADU Configurations Array Entry fields

Field Name	Data Type	# Entries	Description
SendToMID	uint32	1	Outgoing ADU message ID (if packetization is enabled).
NumRecvFromMIDs	uint32	1	Number of valid MIDs in RecvFromMIDs array.
MsgLims	uint32	10	Array of message limits corresponding to each of the message IDs
RecvFromMIDs	uint32	10	Array of MIDs to which the node will subscribe.

5.4. Contact Configuration

This table captures configurations for flowing bundles to a CLA. After loading the table through cFE Table Services, contact configurations in the table can be added or removed from the node with the contact-setup/contact-teardown directives. The table contains the following components:

Table 25. Contact Configuration

Field Name	Data Type	# Entries	Description
contactConfigs	Contact Configs Array	Max contacts TBD	Configurations for each contact (see table below)

The Contact Configurations Array contains the following fields for each entry. The following Convergence Layer Protocols are planned to be supported:

- UDP - User Datagram Protocol
- TCP - Transmission Control Protocol (with a TCP Convergence Layer Protocol (TCPCLP) header)
- EPP - Encapsulation Packet Protocol
- LTP - Licklider Transmission Protocol.
- SB - CCSDS packets over the cFS Software Bus

As of build 7.0.5, only UDP and SB are presently supported.

Table 26. Contact Configurations Array Entry Fields

Field Name	Field Type	Description
destEIDs	EID pattern array of size 3 by default	Destination EID patterns corresponding to this contact
claType	uint32 enum	Type of CLA. UDP (0), TCP (1), EPP (2), LTP (3), SB (4)

Field Name	Field Type	Description
claInAddr	char[10]	CLA ingress IP address
claOutAddr	char[10]	CLA egress IP address
claInPort	uint16	CLA ingress port number
claOutPort	uint16	CLA egress port number
retransmitTimeout	uint32	Bundle reforwarding timeout in milliseconds.
csTimeTrigger	uint32	Custody signal time trigger in milliseconds.
csSizeTrigger	uint32	Custody signal size trigger in bytes.
spare	uint32	Spare bytes for alignment
ingressBitsPerCycle	size_t	Maximum bits to ingress per wakeup cycle. Note: default wakeup rate is 10Hz.
egressBitsPerCycle	size_t	Maximum bits to egress per wakeup cycle. Note: default wakeup rate is 10Hz.

5.5. MIB Node Configuration

This table defines the MIB configurations for the node:

Table 27. MIB Node Configuration

Field ID	Field Name	Field Type	Description
N/A	instanceEID	EID	Endpoint ID of this node
0	<i>paramBundleSizeNoFragment</i>	uint32	Maximum size of bundles that can traverse DTN without additional bundle layer fragmentation (provided by DTNNM).
1	paramSetMaxSequenceNum	uint32	Maximum bundle sequence number
2	paramSetMaxPayloadLength	uint32	Maximum payload length for fragmentation (in bytes)
3	paramSetMaxBundleLength	uint32	Maximum bundle length (in bytes)
4	paramSupportCustody	uint32	Whether to support custody transfer operations (0=false, 1=true)
5	paramSetMaxLifetime	uint32	Maximum bundle lifetime allowed on node (in msec)



Fields with a field ID can be modified via the `set-MIB-item` directive.

5.6. MIB Source Configuration*

This table defines the MIB configurations for each specified source. This table's functionality has not been implemented as of build 7.0.5

Table 28. Source Configuration

Field Name	Data Type	# Entries	Description
SourceConfigs	Source Configs Array	Max sources, 10 by default	Configurations for each source (see table below)

The Source Configurations Array contains the following fields for each entry:

Table 29. Source Configuration Array

Field ID	Field Name	Data Type	# Entries	Description
N/A	SrcEIDs	EID Pattern	Max MIB Per Source EID Patterns, 4 by default	Source EID Patterns
5	ParamSetMaxLifetime	uint32	1	Maximum bundle lifetime (in seconds) that ensures that bundle retention until its expiration time will not degrade operation of the receiving node. Used by node to determine when the bundle must be deleted to prevent network performance degradation.
6	ParamSetMaxBSRGenerationRate	uint32	1	Maximum number of BSRs per minute that can be generated for each source
7	ParamSetMaxCBRGenerationRate	uint32	1	Maximum number of CBRs per minute that can be generated for each source
8	BundleSetBehaviorReceivedBSRGenerate	uint32	1	Flag indicating bundle reception BSRs should be generated
9	BundleSetBehaviorAcceptedBSRGenerate	uint32	1	Flag indicating bundle custody accepted BSRs should be generated

Field ID	Field Name	Data Type	# Entries	Description
10	BundleSetBehaviorForwardedBSRGenerate	uint32	1	Flag indicating bundle forwarded BSRs should be generated
11	BundleSetBehaviorDeliveredBSRGenerate	uint32	1	Flag indicating bundle delivered BSRs should be generated
12	BundleSetBehaviorDeletedBSRGenerate	uint32	1	Flag indicating bundle deleted BSRs should be generated
13	BundleSetBehaviorReceivedCBRGenerate	uint32	1	Flag indicating bundle reception CBRs should be generated
14	BundleSetBehaviorAcceptedCBRGenerate	uint32	1	Flag indicating bundle custody accepted CBRs should be generated
15	BundleSetBehaviorForwardedCBRGenerate	uint32	1	Flag indicating bundle forwarded CBRs should be generated
16	BundleSetBehaviorDeliveredCBRGenerate	uint32	1	Flag indicating bundle delivered CBRs should be generated
17	BundleSetBehaviorDeletedCBRGenerate	uint32	1	Flag indicating bundle deleted CBRs should be generated



Fields with a field ID can be modified via the **set-MIB-item** directive.

5.7. Policy by EID Pattern Authorization Tables*

The Policy by EID Pattern Authorization Tables are: Source Authorization Policy, Custodian Authorization Policy, Report-To-EID Authorization Policy, Source Latency Policy, and Storage Allocation. These tables' functionality have not been implemented as of build 7.0.5.

Note: When a table has only one field its entry is simplified to two columns.

5.7.1. Source Authorization Policy*

This table contains the following components:

authorizedSources

(EID Pattern[10]) Source EIDs from which the node is authorized to receive data.

5.7.2. Custodian Authorization Policy*

This table contains the following components:

authorizedCustodians

(EID Pattern[10]) Custodian block source node IDs to which the node is authorized to acknowledge custody.

5.7.3. Report-To-EID Authorization Policy*

This table contains the following components:

authorizedReportToEIDs

(EID Pattern[10]) Node EIDs to which this node is authorized to send reports.

5.7.4. Source Latency Policy*

This table contains the following components:

Table 30. Latencies

Field Name	Data Type	# Entries	Description
Latencies	Latency Array	Max source latencies, 10 by default	Latencies for each defined source EID (see table below)

The Latency Array contains the following fields for each entry:

sourceEID

(EID Pattern) Source EID pattern

latency

(uint32 enum) Source latency: Low, Medium, High

spare

(uint32) Spare for alignment

5.8. Storage*

The storage allocation defines how to partition the storage database by source EIDs. This table's functionality has not been implemented as of build 7.0.5. This table contains the following components:

Table 31. Partition Configuration

Field Name	Data Type	# Entries	Description
partitionConfigs	Partition Configs Array	Max partitions, 10 by default	Partition sizes for each source EID (see table below)

The Partition Configurations Array contains the following fields for each entry:

sourceEID

(EID Pattern array of length 10 by default) Source EID patterns

partitionSize

(uint32) Storage partition size

Chapter 6. Events

Events are a subset of telemetry with a real-time transport Quality of Service for local fault handling. When a BPNode event occurs, the node logs messages and reports them to operators in the form of the CCSDS space packets.

BPLib_EM_EventType_DEBUG

- (1) Intended only for debugging, not nominal operations

BPLib_EM_EventType_INFORMATION

- (2) Identify a state change or action that is nominal

BPLib_EM_EventType_WARNING

- (3) Identify a state change or action that is not an error but is off-nominal

BPLib_EM_EventType_ERROR

- (4) Identify an error that is not catastrophic (e.g., a bad command)

BPLib_EM_EventType_CRITICAL

- (5) Identify errors that are unrecoverable autonomously.

In a cFS-based system, these event types map onto the equivalent cFS event types, with the exception of BPLib_EM_EventType_WARNING, which maps onto CFE_EVS_EventType_INFORMATION.

The sections below group events by subsystem or module that generates them. All event messages are time-stamped.

6.1. BPNode Main Task Events

These events are within the scope of the cFS BPNode app and are issued by the main task.

Table 32. BPNode Main Task Events

Event Mnemonic	Event ID	Event Type	Description
BPNODE_INIT_INF_EID	1	Informational	Issued after a successful app initialization
BPNODE_CC_ERR_EID	2	Error	An invalid command code was received
BPNODE_NOOP_INF_EID	3	Informational	A no-op command was received
BPNODE_MID_ERR_EID	5	Error	An invalid message ID was received
BPNODE_CMD_LEN_ERR_EID	6	Error	A command with an invalid length was received

Event Mnemonic	Event ID	Event Type	Description
BPNODE_PIPE_ERR_EID	7	Error	A SB pipe error was received when attempting to read a pipe
BPNODE_CR_CMD_PIPE_ERR_EID	8	Error	Error creating the SB command pipe
BPNODE_CR_WKP_PIPE_ERR_EID	9	Error	Error creating the SB wakeup pipe
BPNODE_SUB_CMD_ERR_EID	10	Error	Error subscribing to the SB command pipe
BPNODE_SUB_WKP_ERR_EID	11	Error	Error subscribing to the SB wakeup pipe
BPNODE_TBL_REG_ERR_EID	12	Error	Error registering a table
BPNODE_TBL_LD_ERR_EID	13	Error	Error loading a table
BPNODE_TBL_ADD_R_ERR_EID	14	Error	Error getting the address for a table
BPNODE_TBL_MNG_ERR_EID	15	Error	Error checking for table updates
BPNODE_BPLIB_INIT_ERR_EID	16	Error	Error initializing BPLib
BPNODE_EXIT_CRIT_EID	17	Critical	App is shutting down
BPNODE_AUTO_ADD_APP_INF_EID	18	Informational	Automatically adding and starting an application at initialization
BPNODE_ADU_START_SUB_DBG_EID	19	Debug	Error subscribing to a message ID on a start-application directive
BPNODE_ADU_STOP_UNSUB_DBG_EID	20	Debug	Error unsubscribing to a message ID on a stop-application directive
BPNODE_DEL_HANDLER_ERR_EID	21	Error	Error setting up the delete handler
BPNODE_TIME_WKP_ERR_EID	22	Error	Error performing time maintenance operations
BPNODE_NC_CFG_UPDATE_ERR_EID	24	Error	Error managing configuration updates
BPNODE_INIT_WORK_NOTIF_ERR_EID	25	Error	Error creating the start work child task notification
BPNODE_INIT_INIT_NOTIF_ERR_EID	26	Error	Error creating the initialization child task notification

Event Mnemonic	Event ID	Event Type	Description
BPNODE_INIT_EXIT_NOTIF_ERR_EID	27	Error	Error creating the exit child task notification
BPNODE_INIT_NOTIF_ERR_EID	28	Error	Error waiting on the initialization child task notification
BPNODE_EXIT_NOTIF_ERR_EID	29	Critical	Error waiting on the exit child task notification
BPNODE_TASK_NO_ID_ERR_EID	30	Error	Error getting a child task's ID
BPNODE_TASK_UNK_EXIT_CRIT_EID	31	Critical	Error getting a child task's app data
BPNODE_INIT_STORAGE_NOTIF_ERR_EID	32	Error	Error creating the storage child task notification

6.2. BPNode ADU In Task Events

These events are within the scope of the cFS BPNode app and are issued by the ADU In Tasks. Since there can be multiple ADU In tasks (one for each allowed channel), the channel ID of the task is specified at the beginning of every event message.

Table 33. BPNode ADU In Task Events

Event Mnemonic	Event ID	Event Type	Description
BPNODE_ADU_IN_INIT_INF_EID	50	Informational	Child task initialized
BPNODE_ADU_IN_EXIT_CRIT_EID	51	Critical	Child task is shutting down
BPNODE_ADU_IN_NOTIF_ERR_EID	52	Error	Error pending on start work notification
BPNODE_ADU_IN_INIT_PTR_CRIT_EID	53	Critical	Error getting the child task's channel ID
BPNODE_ADU_IN_MAIN_PTR_CRIT_EID	54	Critical	Error getting the child task's channel ID
BPNODE_ADU_IN_CREATE_PIPE_ERR_EID	55	Error	Error creating ADU in SB pipe
BPNODE_ADU_IN_CREATE_ERR_EID	56	Error	Error creating the child task

6.3. BPNODE ADU Out Task Events

These events are within the scope of the cFS BPNODE app and are issued by the ADU Out Tasks. Since there can be multiple ADU Out tasks (one for each allowed channel), the channel ID of the task is specified at the beginning of every event message.

Table 34. BPNODE ADU Out Task Events

Event Mnemonic	Event ID	Event Type	Description
BPNODE_ADU_OUT_PI_OUT_ERR_EID	23	Error	Error egressing an ADU
BPNODE_ADU_OUT_INIT_INF_EID	60	Informational	Child task initialized
BPNODE_ADU_OUT_EXIT_CRT_EID	61	Critical	Child task is shutting down
BPNODE_ADU_OUT_NOTIF_ERR_EID	62	Error	Error pending on start work notification
BPNODE_ADU_OUT_INIT_PTR_CRT_EID	63	Critical	Error getting the child task's channel ID
BPNODE_ADU_OUT_MAIN_PTR_CRT_EID	64	Critical	Error getting the child task's channel ID
BPNODE_ADU_OUT_CREATE_ERR_EID	76	Error	Error creating the child task

6.4. BPNODE CLA In Task Events

These events are within the scope of the cFS BPNODE app and are issued by the CLA In Tasks. Since there can be multiple CLA In tasks (one for each allowed contact), the contact ID of the task is specified at the beginning of every event message.

Table 35. BPNODE CLA In Task Events

Event Mnemonic	Event ID	Event Type	Description
BPNODE_CLA_IN_INIT_INF_EID	70	Informational	Child task initialized
BPNODE_CLA_IN_NOTIF_ERR_EID	71	Error	Error pending on start work notification
BPNODE_CLA_IN_EXIT_CRT_EID	72	Critical	Child task is shutting down
BPNODE_CLA_IN_INIT_PTR_CRT_EID	73	Critical	Error getting the child task's channel ID

Event Mnemonic	Event ID	Event Type	Description
BPNODE_CLA_IN_M AIN_PTR_CRT_EID	74	Critical	Error getting the child task's channel ID
BPNODE_CLA_IN_FI ND_NAME_ERR_EI D	75	Error	Error finding UDP I/O driver
BPNODE_CLA_IN_C FG_PORT_ERR_EID	76	Error	Error setting UDP port for CLA ingress
BPNODE_CLA_IN_C FG_IP_ERR_EID	77	Error	Error setting IP address for CLA ingress
BPNODE_CLA_IN_C FG_DIR_ERR_EID	78	Error	Error setting I/O direction to input
BPNODE_CLA_IN_C FG_SET_RUN_ERR_ EID	79	Error	Error setting I/O run state for CLA ingress
BPNODE_CLA_IN_C FG_STOP_ERR_EID	80	Error	Error setting I/O run state to stop for CLA ingress
BPNODE_CLA_IN_C REATE_ERR_EID	81	Error	Error creating the child task
BPNODE_CLA_IN_C REATE_PIPE_ERR_E ID	82	Error	Error creating the SB pipe
BPNODE_CLA_IN_S UB_ERR_EID	83	Error	Error subscribing to a message ID
BPNODE_CLA_IN_R ECV_BUF_ERR_EID	84	Error	Error receiving a cFE SB buffer
BPNODE_CLA_IN_I O_READ_ERR_EID	85	Error	Error reading from the UDP port
BPNODE_CLA_IN_U NSUB_ERR_EID	86	Error	Error unsubscribing from a message ID

6.5. BPNode CLA Out Task Events

These events are within the scope of the cFS BPNode app and are issued by the CLA Out Tasks. Since there can be multiple CLA Out tasks (one for each allowed contact), the contact ID of the task is specified at the beginning of every event message.

Table 36. BPNode CLA Out Task Events

Event Mnemonic	Event ID	Event Type	Description
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BPNODE_CLA_OUT_INIT_INF_EID	100	Informational	Child task initialized
BPNODE_CLA_OUT_NOTIF_ERR_EID	101	Error	Error pending on start work notification
BPNODE_CLA_OUT_EXIT_CRT_EID	102	Critical	Child task is shutting down
BPNODE_CLA_OUT_INIT_PTR_CRT_EID	103	Critical	Error getting the child task's channel ID
BPNODE_CLA_OUT_MAIN_PTR_CRT_EID	104	Critical	Error getting the child task's channel ID
BPNODE_CLA_OUT_LIB_LOAD_ERR_EID	105	Error	Error getting bundle for egress
BPNODE_CLA_OUT_FIND_NAME_ERR_EID	106	Error	Error finding UDP I/O driver
BPNODE_CLA_OUT_CFG_PORT_ERR_EID	107	Error	Error setting UDP port for CLA egress
BPNODE_CLA_OUT_CFG_IP_ERR_EID	108	Error	Error setting IP address for CLA egress
BPNODE_CLA_OUT_CFG_DIR_ERR_EID	109	Error	Error setting I/O direction to output
BPNODE_CLA_OUT_CFG_SET_RUN_ERR_EID	110	Error	Error setting I/O run state for CLA egress
BPNODE_CLA_OUT_CFG_STOP_ERR_EID	111	Error	Error setting I/O run state to stop
BPNODE_CLA_OUT_CREATE_ERR_EID	112	Error	Error creating the child task
BPNODE_CLA_OUT_RUN_ERR_EID	113	Error	Error giving/taking the initialization semaphore

6.6. BPNode Generic Worker Task Events

These events are within the scope of the cFS BPNode app and are issued by the Generic Worker tasks. Since there can be multiple Generic Worker tasks, the worker ID of the task is specified at the beginning of every event message.

Table 37. BPNode Generic Worker Events

Event Mnemonic	Event ID	Event Type	Description
BPNODE_GEN_WRKR_INIT_INF_EID	130	Informational	Child task initialized
BPNODE_GEN_WRKR_NOTIF_ERR_EID	131	Error	Error pending on start work notification
BPNODE_GEN_WRKR_EXIT_CRT_EID	132	Critical	Child task is shutting down
BPNODE_GEN_WRKR_INIT_PTR_CRT_EID	133	Critical	Error getting the child task's channel ID
BPNODE_GEN_WRKR_MAIN_PTR_CRT_EID	134	Critical	Error getting the child task's channel ID
BPNODE_GEN_WRKR_TASKRUN_ERR_EID	135	Error	Error running worker job
BPNODE_GEN_WRKR_CREATE_ERR_EID	136	Error	Error creating the child task
BPNODE_GEN_WRKR_REGISTER_ERR_EID	137	Error	Error registering worker with BPLib

6.7. BPNode Generic Worker Task Events

These events are within the scope of the cFS BPNode app and are issued by the Maintenance Task.

Table 38. BPNode Generic Worker Events

Event Mnemonic	Event ID	Event Type	Description
BPNODE_MAINT_INIT_INF_EID	150	Informational	Child task initialized
BPNODE_MAINT_NOTIF_ERR_EID	151	Error	Error pending on start work notification
BPNODE_MAINT_EXIT_CRT_EID	152	Critical	Child task is shutting down
BPNODE_MAINT_CREATE_ERR_EID	153	Error	Error creating the child task
BPNODE_MAINT_PROGRESS_ERR_EID	154	Error	Error loading bundles from storage into memory

6.8. BPLib Node Configuration Events

These events are issued directly by BPLib and relate to directive processing and node configuration.

Table 39. BPLib Node Configuration Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_NC_NOOP_SUCCESS_EID	501	Informational	Success receiving a noop directive
BPLIB_NC_ADD_ALL_APPS_SUCCESS_EID	502	Informational	Success receiving an add-all-applications directive
BPLIB_NC_ADD_APP_SUCCESS_EID	503	Informational	Success receiving an add-application directive
BPLIB_NC_ADD_AUTH_CUSTODIANS_SUCCESS_EID	504	Informational	Success receiving an add-authorized-custodians directive
BPLIB_NC_ADD_AUTH_CUST_SRCS_SUCCESS_EID	505	Informational	Success receiving an add-authorized-custody-sources directive
BPLIB_NC_ADD_AUTH_RPT_EID_SUCCESS_EID	506	Informational	Success receiving an add-authorized-report-to-eid directive
BPLIB_NC_ADD_AUTH_SRCS_SUCCESS_EID	507	Informational	Success receiving an add-authorized-sources directive
BPLIB_NC_ADD_LATENCY_SUCCESS_EID	508	Informational	Success receiving an add-latency directive
BPLIB_NC_ADD_MIB_ARR_KEY_SUCCESS_EID	509	Informational	Success receiving an add-mib-array-key directive
BPLIB_NC_ADD_STORAGE_ALLOC_SUCCESS_EID	510	Informational	Success receiving an add-storage-allocation directive
BPLIB_NC_CLR_VOLATILE_SUCCESS_EID	511	Informational	Success receiving a clear-volatile directive
BPLIB_NC_CONTACT_SETUP_SUCCESS_EID	512	Informational	Success receiving a contact-setup directive

Event Mnemonic	Event ID	Event Type	Description
BPLIB_NC_CONTACT_START_SUCCESS_EID	513	Informational	Success receiving a contact-start directive
BPLIB_NC_CONTACT_STOP_SUCCESS_EID	514	Informational	Success receiving a contact-stop directive
BPLIB_NC_CONTACT_TEARDOWN_SUCCESS_EID	515	Informational	Success receiving a contact-teardown directive
BPLIB_NC_INIT_BUNDLE_STOR_SUCCESS_EID	516	Informational	Success receiving an initialize-bundle-storage directive
BPLIB_NC_PERFORM_SELF_TEST_SUCCESS_EID	517	Informational	Success receiving a perform-self-test directive
BPLIB_NC_REBUILD_BNDL_METADATA_SUCCESS_EID	518	Informational	Success receiving a rebuild-bundle-metadata directive
BPLIB_NC_RELOAD_SVD_DATA_SUCCESS_EID	519	Informational	Success receiving a reload-saved-data directive
BPLIB_NC_RESET_ALL_COUNTERS_SUCCESS_EID	520	Informational	Success receiving a reset-all-counters directive
BPLIB_NC_RESET_BUNDLE_COUNTERS_SUCCESS_EID	521	Informational	Success receiving a reset-bundle-counters directive
BPLIB_NC_RESET_COUNTER_SUCCESS_EID	522	Informational	Success receiving a reset-counter directive
BPLIB_NC_RESET_ERROR_COUNTERS_SUCCESS_EID	523	Informational	Success receiving a reset-error-counters directive
BPLIB_NC_RESET_SOURCE_COUNTERS_SUCCESS_EID	524	Informational	Success receiving a reset-source-counters directive
BPLIB_NC_REMOVE_APP_SUCCESS_EID	525	Informational	Success receiving a remove-application directive
BPLIB_NC_REMOVE_AUTHORIZED_CUSTODIANS_SUCCESS_EID	526	Informational	Success receiving a remove-authorized-custodians directive

Event Mnemonic	Event ID	Event Type	Description
BPLIB_NC_RM_AUT H_CUST_SRCS_SUCC ESS_EID	527	Informational	Success receiving a remove-authorized-custody- sources directive
BPLIB_NC_RM_AUT H_RPT_EID_SUCCES S_EID	528	Informational	Success receiving a remove-authorized-report-to- eid directive
BPLIB_NC_RM_AUT H_SRCS_SUCCESS_E ID	529	Informational	Success receiving a remove-authorized-sources directive
BPLIB_NC_RM_LAT ENCY_SUCCESS_EID	530	Informational	Success receiving a remove-latency directive
BPLIB_NC_RM_MIB _ARR_KEY_SUCCES S_EID	531	Informational	Success receiving a remove-mib-array-key directive
BPLIB_NC_RM_STO R_ALLOC_SUCCESS _EID	532	Informational	Success receiving a remove-storage-allocation directive
BPLIB_NC_SET_MIB _ITEM_SUCCESS_EI D	533	Informational	Success receiving a set-mib-item directive
BPLIB_NC_SET_REG I_STAT_SUCCESS_EI D	534	Informational	Success receiving a set-registration-state directive
BPLIB_NC_START_A LL_APPS_SUCCESS_ EID	535	Informational	Success receiving a start-all-applications directive
BPLIB_NC_START_A PP_SUCCESS_EID	536	Informational	Success receiving a start-application directive
BPLIB_NC_STOP_AP P_SUCCESS_EID	537	Informational	Success receiving a stop-application directive
BPLIB_NC_VERIF_B NDL_META_SUCCE SS_EID	538	Informational	Success receiving a verify-bundle-metadata directive
BPLIB_NC_VERIF_B NDL_SUCCESS_EID	539	Informational	Success receiving a verify-bundle-storage directive
BPLIB_NC_ADD_AL L_APPS_ERR_EID	540	Error	Error receiving an add-all-applications directive
BPLIB_NC_ADD_AP P_ERR_EID	541	Error	Error receiving an add-application directive

Event Mnemonic	Event ID	Event Type	Description
BPLIB_NC_ADD_AUTH_CUSTODIANS_ERR_EID	542	Error	Error receiving an add-authorized-custodians directive
BPLIB_NC_ADD_AUTH_CUST_SRCS_ERR_EID	543	Error	Error receiving an add-authorized-custody-sources directive
BPLIB_NC_ADD_AUTH_RPT_EID_ERR_EID	544	Error	Error receiving an add-authorized-report-to-eid directive
BPLIB_NC_ADD_AUTH_SRCS_ERR_EID	545	Error	Error receiving an add-authorized-sources directive
BPLIB_NC_ADD_LATENCY_ERR_EID	546	Error	Error receiving an add-latency directive
BPLIB_NC_ADD_MIB_ARRAY_KEY_ERR_EID	547	Error	Error receiving an add-mib-array-key directive
BPLIB_NC_ADD_STORAGE_ALLOC_ERR_EID	548	Error	Error receiving an add-storage-allocation directive
BPLIB_NC_CLR_VOLATILE_ERR_EID	549	Error	Error receiving a clear-volatile directive
BPLIB_NC_CONTACT_SETUP_ERR_EID	550	Error	Error receiving a contact-setup directive
BPLIB_NC_CONTACT_START_ERR_EID	551	Error	Error receiving a contact-start directive
BPLIB_NC_CONTACT_STOP_ERR_EID	552	Error	Error receiving a contact-stop directive
BPLIB_NC_CONTACT_TEARDOWN_ERR_EID	553	Error	Error receiving a contact-teardown directive
BPLIB_NC_INIT_BUNDLE_STOR_ERR_EID	554	Error	Error receiving a initialize-bundle-storage directive
BPLIB_NC_PERFORM_SELF_TEST_ERR_EID	555	Error	Error receiving a perform-self-test directive
BPLIB_NC_REBUILD_BNDL_METADATA_ERR_EID	556	Error	Error receiving a rebuild-bundle-metadata directive

Event Mnemonic	Event ID	Event Type	Description
BPLIB_NC_RELOAD_SVD_DATA_ERR_EID	557	Error	Error receiving a reload-saved-data directive
BPLIB_NC_RESET_COUNTER_ERR_EID	558	Error	Error receiving a reset-counter directive
BPLIB_NC_RESET_SOURCE_COUNTERS_ERR_EID	559	Error	Error receiving a reset-source-counters directive
BPLIB_NC_RESET_BUNDLE_COUNTERS_ERR_EID	560	Error	Error receiving a reset-bundle-counters directive
BPLIB_NC_RESET_ERROR_COUNTERS_ERR_EID	561	Error	Error receiving a reset-error-counters directive
BPLIB_NC_RM_APP_ERR_EID	562	Error	Error receiving a remove-application directive
BPLIB_NC_RM_AUTH_CUSTODIANS_ERR_EID	563	Error	Error receiving a remove-authorized-custodians directive
BPLIB_NC_RM_AUTH_CUST_SRCS_ERR_EID	564	Error	Error receiving a remove-authorized-custody-sources directive
BPLIB_NC_RM_AUTH_RPT_EID_ERR_EID	565	Error	Error receiving a remove-authorized-report-to-eid directive
BPLIB_NC_RM_AUTH_SRCS_ERR_EID	566	Error	Error receiving a remove-authorized-sources directive
BPLIB_NC_RM_LATENCY_ERR_EID	567	Error	Error receiving a remove-latency directive
BPLIB_NC_RM_MIB_ARR_KEY_ERR_EID	568	Error	Error receiving a remove-mib-array-key directive
BPLIB_NC_RM_STORAGE_ALLOC_ERR_EID	569	Error	Error receiving a remove-storage-allocation directive
BPLIB_NC_SET_MIB_ITEM_ERR_EID	570	Error	Error receiving a set-mib-item directive
BPLIB_NC_SET_REGISTRATION_STATE_ERR_EID	571	Error	Error receiving a set-registration-state directive
BPLIB_NC_START_ALL_APPS_ERR_EID	572	Error	Error receiving a start-all-applications directive
BPLIB_NC_START_APPLICATION_ERR_EID	573	Error	Error receiving a start-application directive

Event Mnemonic	Event ID	Event Type	Description
BPLIB_NC_STOP_APP_ERR_EID	574	Error	Error receiving a stop-application directive
BPLIB_NC_VERIFY_BNDL_ERR_EID	575	Error	Error receiving a verify-bundle-storage directive
BPLIB_NC_VERIFY_BNDL_META_ERR_EID	576	Error	Error receiving a verify-bundle-metadata directive
BPLIB_NC_SEND_NODE_CONFIG_ERR_EID	577	Error	Error receiving a send-node-mib-config-hk directive
BPLIB_NC_SEND_SOURCE_CONFIG_ERR_EID	578	Error	Error receiving a send-source-mib-config-hk directive
BPLIB_NC_SEND_NODE_CNTRS_ERR_EID	579	Error	Error receiving a send-node-mib-counters-hk directive
BPLIB_NC_SEND_SOURCE_CNTRS_ERR_EID	580	Error	Error receiving a send-source-mib-counters-hk directive
BPLIB_NC_SEND_STORAGE_ERR_EID	581	Error	Error receiving a send-storage-hk directive
BPLIB_NC_SEND_CONTACTS_ERR_EID	582	Error	Error receiving a send-channel-contact-status-hk directive
BPLIB_NC_SEND_REPORTS_ERR_EID	583	Error	Error receiving a send-node-mib-reports-hk directive
BPLIB_NC_STORAGE_CLEANUP_INF_EID	584	Informational	Success receiving a clean-storage directive
BPLIB_NC_STORAGE_CLEANUP_ERR_EID	585	Error	Error receiving a clean-storage directive
BPLIB_NC_MAINT_ERR_EID	586	Error	Error performing node maintenance activities
BPLIB_NC_TBL_UPDATE_INF_EID	587	Informational	Updated a configuration table successfully
BPLIB_NC_INIT_ERR_EID	589	Error	Error initializing the node

6.9. BPLib Admin Statistics (AS) Events

These events are issued directly by BPLib and relate to AS operations.

Table 40. BPLib Admin Statistics Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_AS_TAKE_M UTEX_ERR_EID	590	Error	Error taking counters mutex
BPLIB_AS_GIVE_M UTEX_ERR_EID	591	Error	Error giving counters mutex
BPLIB_AS_SET_CTR _ERR_EID	592	Error	Error incrementing/decrementing a counter

6.10. BPLib Storage Events

These events are issued directly by BPLib and relate to Storage operations.

Table 41. BPLib Storage Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_STOR_SQL_L OAD_ERR_EID	608	Error	Error loading bundles from storage for egress
BPLIB_STOR_SQL_S TORE_ERR_EID	609	Error	Error storing bundle
BPLIB_STOR_SQL_G C_ERR_EID	610	Error	Error discarding egressed/expired bundles
BPLIB_STOR_DB_FU LL_INF_EID	611	Informational	Storage is full, dropping bundles
BPLIB_STOR_DB_G ET_SIZE_ERR_EID	612	Error	Error getting database size
BPLIB_STOR_EXPIR E_DBG_EID	613	Debug	Reports the number of expired bundle discarded from storage
BPLIB_STOR_DELE TE_DBG_EID	614	Debug	Reports the number of egressed bundles discarded from storage
BPLIB_STOR_DUPL_ DBG_EID	615	Debug	Reports the number of duplicate bundle found in a storage batch
BPLIB_STOR_CLEA NUP_INF_EID	616	Informational	Informs the user that storage cleanup has started
BPLIB_STOR_CCS_E RR_EID	617	Error	Error stating that something went wrong while attempting to store a CCS
BPLIB_STOR_RETR ANSMIT_UPDATE_ DBG_EID	618	Debug	Reports that retransmit triggers for bundles for the provided contact were updated

Event Mnemonic	Event ID	Event Type	Description
BPLIB_STOR_AUTO_CLEAN_START_INF_EID	619	Informational	Informs the user that automatic storage cleanup has started
BPLIB_STOR_AUTO_CLEAN_END_INF_EID	620	Informational	Informs the user that automatic storage cleanup has completed
BPLIB_STOR_AUTO_CLEAN_ERR_EID	621	Error	Error, something went wrong while trying to clean up storage
BPLIB_STOR_NO_MEM_ERR_EID	622	Error	No more memory is available when trying to load a bundle from storage
BPLIB_STOR_UNKOWN_CUST_ERR_EID	623	Error	Mismatch detected between number of custodial bundles stored and number of custodial bundles in storage batch

6.11. BPLib Bundle Interface Events

These events are issued directly by BPLib and relate to BI and CLA operations.

Table 42. BPLib BI/CLA Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_BI_INGRESS_CBOR_DECODE_INF_EID	650	Informational	Error ingressing a bundle
BPLIB_CLA_CONTACT_NO_STATE_CHG_DBG_EID	661	Debug	A contact directive was received that didn't change the contact state
BPLIB_CLA_INVALID_CONTACT_ID_DBG_EID	662	Debug	Invalid contact ID provided by a contact directive
BPLIB_CLA_REMOVE_QUEUE_FLUSH_DBG_EID	663	Debug	Error storing bundles in egress queue upon receiving a contact-teardown directive
BPLIB_CLA_DUPLICATE_EIDS_WRN_EID	664	Warning	Duplicate destination EIDs detected in a contact table
BPLIB_CLA_RESET_TRIGGERS_DBG_EID	665	Debug	Information regarding the retransmission triggers that were reset upon a contact-setup directive

6.12. BPLib Payload Interface (PI) Events

These events are issued directly by BPLib and relate to PI operations.

Table 43. BPLib PI Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_PI_EGRESS_ERR_EID	682	Error	Error copying ADU out for egress
BPLIB_PI_INGRESS_ERR_EID	683	Error	Error ingressing an ADU
BPLIB_PI_REMOVE_STATE_DBG_EID	684	Debug	Invalid app state upon receiving a remove-application directive
BPLIB_PI_REMOVE_ID_DBG_EID	685	Debug	Invalid channel ID in a remove-application directive
BPLIB_PI_REMOVE_FWP_DBG_EID	686	Debug	Framework-specific error upon receiving a remove-application directive
BPLIB_PI_ADD_ID_DBG_EID	687	Debug	Invalid channel ID in an add-application directive
BPLIB_PI_ADD_STATE_DBG_EID	688	Debug	Invalid app state upon receiving an add-application directive
BPLIB_PI_ADD_FWP_DBG_EID	689	Debug	Framework-specific error upon receiving an add-application directive
BPLIB_PI_START_ID_DBG_EID	690	Debug	Invalid channel ID in a start-application directive
BPLIB_PI_START_STATE_DBG_EID	691	Debug	Invalid app state upon receiving a start-application directive
BPLIB_PI_START_FWP_DBG_EID	692	Debug	Framework-specific error upon receiving a start-application directive
BPLIB_PI_STOP_ID_DBG_EID	693	Debug	Invalid channel ID in a stop-application directive
BPLIB_PI_STOP_STATE_DBG_EID	694	Debug	Invalid app state upon receiving a stop-application directive
BPLIB_PI_STOP_FWP_DBG_EID	695	Debug	Framework-specific error upon receiving a stop-application directive
BPLIB_PI_REMOVE_QUEUE_FLUSH_DBG_EID	696	Debug	Error storing bundles in egress queue upon receiving a remove-application directive
BPLIB_PI_ADU_LEN_ERR_EID	697	Error	Error ingressing an ADU, invalid ADU length

Event Mnemonic	Event ID	Event Type	Description
BPLIB_PI_INGRESS_NO_STOR_ERR_EID	698	Error	Error ingressing an ADU, no storage available
BPLIB_PI_NO_CTEB_DBG_EID	699	Error	Error with add-application directive CTEB configuration

6.13. BPLib Queue Management (QM) Events

These events are issued directly by BPLib and relate to queue and job operations.

Table 44. BPLib QM Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_QM_EBP_OUT_ERR_EID	700	Error	Error updating extension blocks
BPLIB_QM_CT_IN_ERR_EID	701	Error	Error processing custody transfer for new bundles
BPLIB_QM_CT_OUT_ERR_EID	702	Error	Error occurred processing custody transfer for bundles being delivered

6.14. BPLib Custody Transfer (CT) Events

These events are issued directly by BPLib and relate to custody transfer operations.

Table 45. BPLib CT Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_CT_INV_SEQ_NUM_DBG_EID	752	Debug	Invalid sequence number detected in CCS bundle sequence range
BPLIB_CT_NONCUSTODIAL_ERR_EID	753	Error	CT received an egressing bundle whose custody has not been finalized
BPLIB_CT_CCS_CRRPTD_ERR_EID	754	Error	CCS data in memory failed the sanity checks of the sequence range length
BPLIB_CT_INV_EGRESS_ID_ERR_EID	755	Error	The channel a custodial bundle should be egressed on is invalid
BPLIB_CT_NO_STORE_ERR_EID	756	Error	The size of the bundle-to-be-stored exceeds the remaining space in storage
BPLIB_CT_NO_MEM_ERR_EID	757	Error	The CTDB doesn't have remaining entries for the incoming custodial

Event Mnemonic	Event ID	Event Type	Description
BPLIB_CT_REJECTED_DBG_EID	758	Debug	Custody for 1 or more bundles was rejected by a downstream node
BPLIB_CT_UNKNOW_BUNDLE_WRN_EID	759	Warning	CTDB is corrupted or the node was restarted with custodial bundles in storage
BPLIB_CT_DB_DELETE_ERR_EID	760	Error	A custodial bundle could not be deleted from the CTDB
BPLIB_CT_CCS_INSERT_ERR_EID	761	Error	Unknown error occurred when trying to insert a bundle into a CCS

6.15. BPLib Admin Record Processor (ARP) Events

These events are issued directly by BPLib and relate to administrative record operations.

Table 46. BPLib ARP Events

Event Mnemonic	Event ID	Event Type	Description
BPLIB_ARP_CREATE_JOB_ERR_EID	800	Error	Failed to put the new administrative record bundle on the job queue
BPLIB_ARP_NULL_BUNDLE_ERR_EID	801	Error	Unable to allocate memory for a new administrative record bundle

Appendix A: Acronyms and Abbreviations

Acronym or Abbreviation	Description
ADU	Application Data Unit
AS	Admin Statistics
BI	Bundle Interface
BP	Bundle Protocol
BPNode	Bundle Protocol Node
BPLib	Bundle Protocol Library
BSR	Bundle Status Report
CBOR	Concise Binary Object Representation
CBR	Compressed Bundle Reporting
CCB	Configuration Control Board
CS	Custody Signal
CCS	Compressed Custody Signal
CCSDS	Consultative Committee for Space Data Systems
cFS	core Flight System
CL	Convergence Layer
CLA	Convergence Layer Adapter
CM	Configuration Management
CRC	Cyclic Redundancy Check
CREB	Compressed Reporting Extension Block
CRS	Compressed Reporting Signal
CT	Custody Transfer
CTDB	Custody Transfer Database
CTEB	Custody Transfer Extension Block
CTI	Custody Transfer Items
DOORS	Dynamic Object-Oriented Requirements System
DTN	Delay/Disruption-Tolerant Networking
DTNNM	DTN Network Management
EID	Endpoint Identifier
FSW	Flight Software
GSFC	Goddard Space Flight Center
HK	Housekeeping

Acronym or Abbreviation	Description
ICD	Interface Control Document
IETF	Internet Engineering Task Force
LTP	Licklider Transmission Protocol
M&C	Monitor and Control
MIB	Management Information Base
MOC	Mission Operations Center
NC	Node Configuration
PI	Payload Interface
QM	Queue Management
RVTM	Requirements Verification Traceability Matrix
SB	Software Bus
SLA	Service Level Agreement
TDMS	Technical Data Management System